

PROJECT REPORT

HR Analytics

Employee Attrition & Salary Prediction using Machine Learning

An interactive Streamlit dashboard and FastAPI service that analyses workforce data to predict attrition risk and estimate salary using classification and regression models.

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Python · Streamlit · FastAPI · Scikit-learn · MySQL

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About the Company



Excellence Technology

Excellence Technology is a leading IT training and software development company based in Mohali, Punjab. Established in 2013, the company specializes in providing industry-oriented training programs, software development services, and internship opportunities for students and professionals. The organization focuses on bridging the gap between academic learning and industry requirements through practical, project-based training.



Internship and Training Programs

Excellence Technology provides industrial training and internship programs for BCA, MCA, B.Tech, Diploma, and MBA students. The training includes hands-on experience through live projects, practical assignments, and mentorship from industry professionals. The objective is to help students develop technical skills and gain real-world exposure to the IT industry.



Vision

To empower students and professionals with industry-relevant skills, practical knowledge, and innovative solutions that contribute to career growth and technological advancement.

Key Highlights

- Established in 2013
- Based in Mohali, Punjab
- Industry-focused practical training
- Live project experience
- Internship and placement assistance
- Multiple training centers across North India
- Focus on emerging technologies and career development

Services Offered

- Python Programming
- Artificial Intelligence & Machine Learning
- Data Science & Analytics
- Full Stack Web Development
- Mobile App Development
- Cybersecurity & Ethical Hacking
- Cloud Computing
- Digital Marketing
- Graphic Designing

About the Project

Introduction

Choosing the right people decisions — who might leave, and what to pay — is one of the hardest, highest-stakes jobs in HR.

The HR Analytics platform gives HR teams a data-driven way to answer both questions: it studies an employee's profile — role, tenure, satisfaction, overtime, income and more — and predicts attrition risk and a fair salary estimate.

The system is built as a **FastAPI REST backend** serving trained Scikit-learn models, paired with an interactive **Streamlit dashboard** for exploration and live predictions, backed by a **MySQL** employee database.



Dataset: IBM HR Analytics Employee Attrition data — 35 features covering demographics, job role, satisfaction, income and tenure.



Employee Data



Preprocessing & Encoding



ML Models (Train / Predict)



Insights & Predictions

Objectives



Predict employee attrition risk using classification models



Estimate fair monthly salary using regression



Store and retrieve employee data via a MySQL database



Visualise workforce trends through EDA charts



Expose predictions through a documented REST API



Help HR teams make faster, evidence-based decisions

Features



Employee Records

List, view and add employees via API



Attrition Prediction

Logistic Regression, KNN, Decision Tree, Random Forest



Salary Prediction

Linear Regression on job & tenure features



Model Performance

Accuracy, confusion matrix, R^2 score



EDA Visualizations

Attrition by department, age, overtime, income



MySQL Integration

Persistent employee data store



REST API

FastAPI endpoints mirroring the dashboard



Streamlit Dashboard

Interactive, no-code exploration UI

Technology Stack



Frontend

- Streamlit



Backend

- Python
- FastAPI



Database

- MySQL
- mysql-connector-python

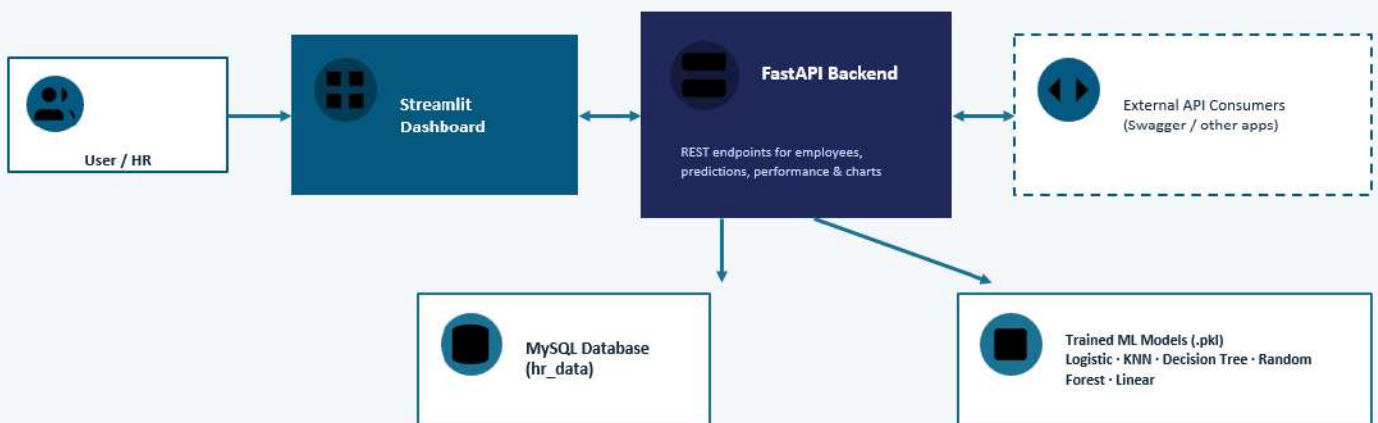


Machine Learning

- Scikit-learn
- Decision Tree · Random Forest
- KNN · Logistic & Linear Regression

Supporting libraries: Pandas · NumPy · Matplotlib · Seaborn · Joblib

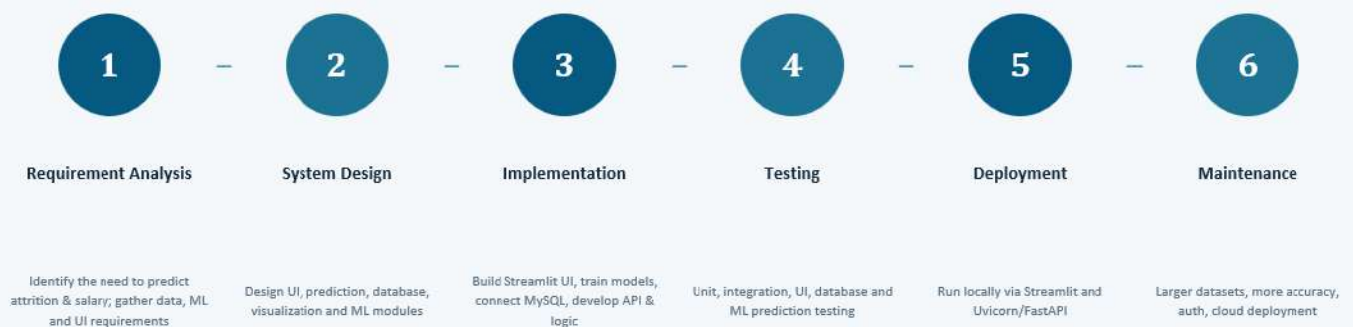
System Architecture



The FastAPI backend mirrors every capability of the Streamlit dashboard as documented REST endpoints, so any client can call the same models and data.

SDLC

Software Development Life Cycle



Hardware Requirements



Developer Side

- Intel Core i3 / i5 Processor or above
- 8 GB RAM
- 256 GB SSD
- Windows 10 / 11



User Side

- Dual-core processor
- 4 GB RAM
- Internet connection
- Windows / Linux / macOS

Software Requirements



Developer Side

- Python 3.x
- VS Code
- Streamlit & FastAPI (Uvicorn)
- MySQL & MySQL Workbench
- Git (optional)



User Side

- Google Chrome
- Microsoft Edge
- Mozilla Firefox
- Streamlit runtime

Machine Learning Approach

Data Pipeline

- 1 Load employee data from MySQL
- 2 Clean missing values & drop identifier columns
- 3 Label-encode categorical fields
- 4 Standard-scale numerical features
- 5 Train / test split (80 / 20)



Random Forest feature importance highlights the strongest drivers of attrition, exposed via the /model-performance endpoint.

Models & Targets



Attrition (classification)

Logistic Regression · KNN · Decision Tree · Random Forest



Salary (regression)

Linear Regression on job, tenure & performance features

Evaluation Metrics

Classification:

Accuracy score & confusion matrix for every model, compared side by side

Regression:

R² score on held-out salary data

Code Highlights

Backend/main.py — FastAPI prediction endpoints

```
app.py

import streamlit as st
import joblib
import pandas as pd
import os
import sys
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import accuracy_score, confusion_matrix, r2_score

st.set_page_config(page_title="HR Analytics Dashboard", layout="wide")

st.title("HR Analytics · Attrition & Salary Prediction")
st.write("Explore workforce data, predict attrition risk, estimate salary, and check model performance.")
```

```

BASE_DIR = os.path.dirname(os.path.abspath(__file__))
MODELS_DIR = os.path.abspath(os.path.join(BASE_DIR, "..", "models"))

# Force absolute path resolution for the backend source directory
SRC_DIR = os.path.abspath(os.path.join(BASE_DIR, "..", "src"))
BACKEND_DIR = os.path.abspath(os.path.join(BASE_DIR, "..", "Backend"))

# Append both possibilities to the system path safely
if SRC_DIR not in sys.path:
    sys.path.append(SRC_DIR)
if BACKEND_DIR not in sys.path:
    sys.path.append(BACKEND_DIR)

from db_connection import get_employee_data, insert_employee
from preprocessing import clean_data, get_attrition_data, get_salary_data

# -----
# CACHED DATA & ARTIFACT LOADERS
# -----

```

```

@st.cache_resource
def load_ml_artifacts():
    log_model = joblib.load(os.path.join(MODELS_DIR, "logistic_model.pkl"))
    knn_model = joblib.load(os.path.join(MODELS_DIR, "knn_model.pkl"))
    dt_model = joblib.load(os.path.join(MODELS_DIR, "decision_tree_model.pkl"))
    rf_model = joblib.load(os.path.join(MODELS_DIR, "random_forest_model.pkl"))
    attrition_encoders = joblib.load(os.path.join(MODELS_DIR, "encoder_log.pkl"))
    attrition_scaler = joblib.load(os.path.join(MODELS_DIR, "scaler_log.pkl"))

    lin_model = joblib.load(os.path.join(MODELS_DIR, "Linear_model.pkl"))
    salary_encoders = joblib.load(os.path.join(MODELS_DIR, "encoder_lin.pkl"))
    salary_scaler = joblib.load(os.path.join(MODELS_DIR, "scaler_lin.pkl"))

    return (log_model, knn_model, dt_model, rf_model, attrition_encoders, attrition_scaler,
            lin_model, salary_encoders, salary_scaler)

(log_model, knn_model, dt_model, rf_model, attrition_encoders, attrition_scaler,
lin_model, salary_encoders, salary_scaler) = load_ml_artifacts()

```

data.csv

Age,Attrition,BusinessTravel,DailyRate,Department,DistanceFromHome,Education,EducationField,**EmployeeCount**,EmployeeNumber,EnvironmentSatisfaction,Gender,HourlyRate,JobInvolvement,JobLevel,JobRole,JobSatisfaction,MaritalStatus,**MonthlyIncome**,MonthlyRate,NumCompaniesWorked,Over18,OverTime,PercentSalaryHike,PerformanceRating,RelationshipSatisfaction,StandardHours,StockOptionLevel,**TotalWorkingYears**,TrainingTimesLastYear,WorkLifeBalance,YearsAtCompany,YearsInCurrentRole,YearsSinceLastPromotion,YearsWithCurrManager

41,Yes,Travel_Rarely,1102,Sales,1,2,Life Sciences,1,1,2,Female,94,3,2,Sales Executive,4,Single,**5993**,19479,8,Y,Yes,11,3,1,80,0,8,0,1,6,4,0,5

49,No,Travel_Frequently,279,Research & Development,8,1,Life Sciences,1,2,3,Male,61,2,2,Research Scientist,2,Married,**5130**,24907,1,Y,No,23,4,4,80,1,**10**,3,3,10,7,1,7

37,Yes,Travel_Rarely,1373,Research & Development,2,2,Other,1,4,4,Male,92,2,1,Laboratory Technician,3,Single,**2090**,2396,6,Y,Yes,15,3,2,80,0,**7**,3,3,0,0,0,0

33,No,Travel_Frequently,1392,Research & Development,3,4,Life Sciences,1,5,4,Female,56,3,1,Research Scientist,3,Married,**2909**,23159,1,Y,Yes,11,3,3,80,0,**8**,3,3,8,7,3,0

Code Highlights

Backend/main.py — FastAPI prediction endpoints

```
main.py

import os
import sys

from fastapi import FastAPI, HTTPException
from fastapi.responses import StreamingResponse
from fastapi.middleware.cors import CORSMiddleware

from schemas import (
    EmployeeIn,
    EmployeeAddedResponse,
    EmployeeListResponse,
    AttritionPredictionRequest,
    AttritionPredictionResponse,
    SalaryPredictionRequest,
    SalaryPredictionResponse,
```

```

ModelPerformanceResponse,
)
from ml_utils import (
    load_resources,
    refresh_employee_data,
    predict_attrition,
    predict_salary,
    get_model_performance,
    generate_visualization,
    VALID_GRAPH_TYPES,
)
from db_connection import insert_employee

app = FastAPI(
    title="HR Analytics API",
    description=(
        "REST API mirroring the HR Analytics Streamlit dashboard: employee "
        "data, attrition/salary predictions, model performance, and visualizations."
    ),
    version="1.0.0",
)

```

OUTPUT

Dashboard

Internal HR analytics tool

Total employees

1473

Historical attrition rate

16.2%

Models Active: Logistic Regression - KNN - Decision Tree - Random Forest - Linear Regression

HR Analytics · Attrition & Salary Prediction

Explore workforce data, predict attrition risk, estimate salary, and check model performance.

Data Explorer

Attrition Predictor (Logistic/KNN)

Decision Tree & Random Forest

Salary Predictor

Model Performance

Visualizations

Add Employee

Data Explorer

Browse the full cleaned employee dataset below.

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EnvironmentSatisfaction	Gender	HourlyRate	JobInvolvement	JobLevel	JobRole	JobSatisfaction	MaritalStatus	Mon
0	41	Yes	Travel_Rarely	1102	Sales		1	2 Life Sciences		2 Female	94	3	2	Sales Executive	4	Single	
1	49	No	Travel_Frequently	279	Research & Development		8	1 Life Sciences		3 Male	61	2	2	Research Scientist	2	Married	
2	37	Yes	Travel_Rarely	1373	Research & Development		2	2 Other		4 Male	92	2	1	Laboratory Technician	3	Single	
3	33	No	Travel_Frequently	1382	Research & Development		3	4 Life Sciences		4 Female	56	3	1	Research Scientist	3	Married	
4	27	No	Travel_Rarely	591	Research & Development		2	1 Medical		1 Male	40	3	1	Laboratory Technician	2	Married	
5	32	No	Travel_Frequently	1005	Research & Development		2	2 Life Sciences		4 Male	79	3	1	Laboratory Technician	4	Single	
6	59	No	Travel_Rarely	1324	Research & Development		3	3 Medical		3 Female	81	4	1	Laboratory Technician	1	Married	
7	30	No	Travel_Rarely	1358	Research & Development		24	1 Life Sciences		4 Male	67	3	1	Laboratory Technician	3	Divorced	
8	38	No	Travel_Frequently	216	Research & Development		23	3 Life Sciences		4 Male	44	2	3	Manufacturing Director	3	Single	
9	36	No	Travel_Rarely	1299	Research & Development		27	3 Medical		3 Male	94	3	2	Healthcare Representati	3	Married	

HR Analytics · Attrition & Salary Prediction

Explore workforce data, predict attrition risk, estimate salary, and check model performance.

[Data Explorer](#) [Attrition Predictor \(Logistic/KNN\)](#) [Decision Tree & Random Forest](#) [Salary Predictor](#) [Model Performance](#) [Visualizations](#) [Add Employee](#)

Attrition Predictor (Logistic Regression / KNN)

Choose model

Logistic Regression

Logistic Regression

KNN

OverTime

Yes

Job Satisfaction (1-4)

3

Years at Company

3

Predict Attrition

(Logistic Regression) This employee is likely to stay. (Probability of leaving: 15.60%)

HR Analytics · Attrition & Salary Prediction

Explore workforce data, predict attrition risk, estimate salary, and check model performance.

Data Explorer Attrition Predictor (Logistic/KNN) **Decision Tree & Random Forest** Salary Predictor Model Performance Visualizations Add Employee

Decision Tree & Random Forest Predictor

Age

30

Monthly Income

5000

-

+

Total Working Years

5

OverTime

Yes

▼

Job Satisfaction (1-4)

3

Years at Company

3

Predict Attrition

Decision Tree

Likely to leave (Probability: 81.31%)

Random Forest

Likely to stay (Probability of leaving: 25.47%)

HR Analytics - Attrition & Salary Prediction

Explore workforce data, predict attrition risk, estimate salary, and check model performance.

Data Explorer Attrition Predictor (Logistic/KNN) Decision Tree & Random Forest Salary Predictor Model Performance Visualizations Add Employee

Model Performance

Logistic Accuracy

87.5%

KNN Accuracy

82.0%

Decision Tree Accuracy

69.8%

Random Forest Accuracy

84.1%

Salary ^{R^2} Score

0.919

Confusion matrix for

Logistic Regression

Confusion Matrix (Logistic Regression)

	Predict: Stay	Predict: Leave
Actual: Stay	208	9
Actual: Leave	28	29

Model Comparison

Model	Accuracy
0 Logistic Regression	0.8750
1 KNN	0.8200
2 Decision Tree	0.6983
3 Random Forest	0.8417



Random Forest - Top Feature Importances



Application Modules & API Endpoints

FastAPI routes mirroring the Streamlit dashboard

Method	Endpoint	Description
GET	/employees	List cleaned employee records (paginated)
GET	/employees/{id}	Fetch a single employee by number
POST	/employees	Add a new employee record to MySQL
POST	/predict/attrition	Predict Stay / Leave with probability
POST	/predict/salary	Predict monthly income (regression)
GET	/model-performance	Accuracy, confusion matrix, feature importance
GET	/visualizations/{type}	PNG chart: department, age, overtime, income, satisfaction, correlation

API OUTPUT

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8000/visualizations/age_distribution' \
-H 'accept: application/json'
```

Request URL

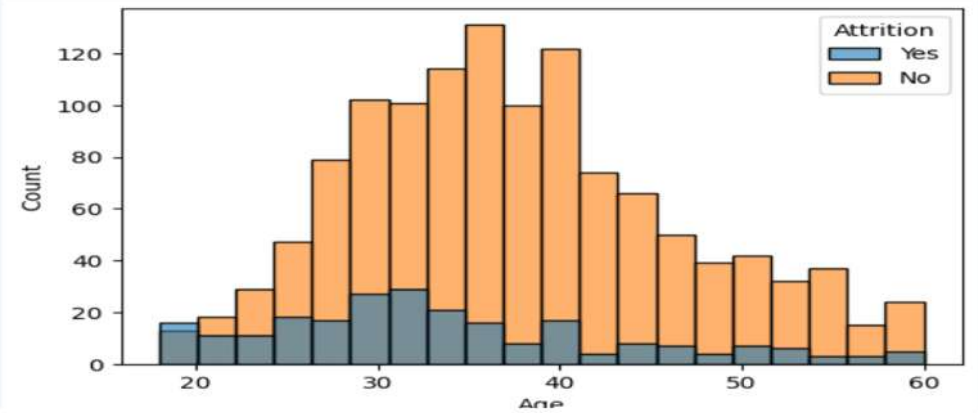
`http://127.0.0.1:8000/visualizations/age_distribution`

Server response

Code Details

200

Response body



```
'http://127.0.0.1:8000/predict/salary' \
-H 'accept: application/json' \
-H 'Content-Type: application/json' \
-d '{
  "age": 18,
  "education": 1,
  "job_level": 1,
  "job_role": "Sales Executive",
  "total_working_years": 40,
  "years_at_company": 40
}'
```

Request URL

http://127.0.0.1:8000/predict/salary

Server response

Code	Details
------	---------

200	
-----	--

Response body

```
{
  "predicted_monthly_income": 5031.434455112181
}
```

Response headers

```
access-control-allow-origin: *
content-length: 46
content-type: application/json
date: Sun, 12 Jul 2026 11:26:28 GMT
server: uvicorn
```

Responses

Code	Description
------	-------------

200	
-----	--

Successful Response

Conclusion

The HR Analytics platform shows how Machine Learning can support real HR decisions — flagging attrition risk and estimating salary directly from an employee's profile. A FastAPI backend and Streamlit dashboard work off the same trained models and MySQL data, so the same predictions are available both as a live dashboard and as a documented REST API for other systems to call.

Future Scope

- Larger, more diverse workforce datasets
- Higher prediction accuracy & model tuning
- User authentication & role-based access
- Cloud deployment of API & dashboard
- Resume analysis & skill recommendations



Attrition Prediction



Salary Estimation



REST API



Interactive Dashboard

Thank You